BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Validation and Authentication

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. After correcting an assessment answer, why attempt a new question?

2. After one ascending bubble-sort pass on [7, 3, 6, 2], what is the list?

A) [2, 3, 6, 7]
B) [3, 6, 2, 7]
C) [7, 3, 6, 2]
D) [2, 6, 3, 7]

3. "An IF statement must execute every possible branch regardless of its condition." Correct this. Explain why.

4. A circuit has X = A AND B, followed by Q = X OR C. For input rows 000, 011, 110, 111, give the pairs (X,Q). Why is a column for X useful?

Today's objectives

Explain the core ideas of validation and authentication using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Validation

Checks whether input meets a defined rule, not whether it is factually true.

Range check

Rejects a number outside stated minimum and maximum boundaries.

Presence and type

Detects missing required input or wrong type/format.

Authentication

Checks a user's claimed identity, which differs from validating an entered value.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Validation	Checks whether input meets a defined rule, not whether it is factually true.
Range check	Rejects a number outside stated minimum and maximum boundaries.
Presence and type	Detects missing required input or wrong type/format.
Authentication	Checks a user's claimed identity, which differs from validating an entered value.

I Do - One worked example

Design a range check for an integer mark from 0 to 100 inclusive.

1. Input the candidate mark.
2. Test mark >= 0 AND mark <= 100.
3. Accept only when both conditions are true.
4. Otherwise display an error and ask for a new input.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A form accepts ages from 13 to 18. Write an acceptance condition.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Finish the inclusive range check for one IPv4 octet.

```
octet = 255
if ... then
print("Valid range")
else
print("Invalid")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. Explain validation versus verification. [2 marks]

Q3. Give a suitable boundary value when valid ages are 13-18. [2 marks]

Q4. State one difference between authentication and validation. [2 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Validation proves a pupil's entered date is factually correct. ☐ T ☐ F
2. Checks whether input meets a defined rule, not whether it is factually true. ☐ T ☐ F
3. Rejects a number outside stated minimum and maximum boundaries. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 To check that the method can be applied independently rather than copied. ☐

Q2 B - [3, 6, 2, 7] ☐

Q3 It chooses a branch using a condition. Selection can be nested. ☐

Q4 (a) (0,0), (0,1), (1,1), (1,1). X shows the first gate's output. This helps work out Q and find errors. ☐

Code First - Model

```
octet = 255
if octet >= 0 AND octet <= 255 then
print("Valid range")
else
print("Invalid")
endif
```

Output: Valid range

We Do - Check the model

1. age >= 13 AND age <= 18. ☐

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Validation checks a rule or format; verification checks data matches the original/source input. ☐

Q3 13 or 18, plus an adjacent invalid such as 12 or 19 for a test set. ☐

Q4 Authentication confirms an asserted identity; validation checks input against constraints. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: